STACKS, DERIVED CATEGORIES OF SPACES: BASE-CHANGE-TENSOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5624-5636 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 5638-5672, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 34 of queue_spaceperfect.json (pb=1183, stmt 5624-5636, proof 5638-5672).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-base-change-tensor' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DKH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5488–5505. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DKG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5365–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DKI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5567–5592. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of algebraic spaces over S. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $\mathcal{G}^\bullet$ be a bounded above complex of quasi-coherent $\mathcal{O}_X$-modules flat over Y. Then formation of*

$$Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet)$$

commutes with arbitrary base change (see proof for precise statement).

Proof. The statement means the following. Let $g : Y' \to Y$ be a morphism of algebraic spaces and consider the base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

in other words $X' = Y' \times_Y X$. The lemma asserts that

$$Lg^* Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet) \longrightarrow Rf'_*(L(g')^* E \otimes^{\mathbf{L}}_{\mathcal{O}_{X'}} (g')^* \mathcal{G}^\bullet)$$

is an isomorphism. Observe that on the right hand side we do **not** use derived pullback on $\mathcal{G}^\bullet$. To prove this, we apply Lemmas 0DKH and 0DKI to see that it suffices to prove the canonical map

$$L(g')^* \mathcal{G}^\bullet \to (g')^* \mathcal{G}^\bullet$$

satisfies the equivalent conditions of Lemma 0DKG. This follows by checking the condition on stalks, where it immediately follows from the fact that $\mathcal{G}^\bullet_{\overline{x}} \otimes_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y',\overline{y}'}$ computes the derived tensor product by our assumptions on the complex $\mathcal{G}^\bullet$. $\square$